AN OPEN MAP WITH IRREDUCIBLE FIBRES INDUCES A BIJECTION ON IRREDUCIBLE COMPONENTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that an irreducible component is closed and that the image of an irreducible subset is irreducible. Invoked inside the counted proof (source lines 1367-1370). Makes components closed and their images irreducible.
- The statement that the restriction of an open map with irreducible fibres to the preimage of an irreducible subset again has irreducible fibres. Invoked near the end of the counted proof (source lines 1371-1372). Shows the preimage of a component is irreducible.

Source reference. Original definition definition-irreducible-components: sources/stacks-topology/pinned-original-source.tex, source lines 1072–1072. Navigation pointer only: the counted text refers to the Stacks item definition-irreducible-components, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-image-irreducible-space: sources/stacks-topology/pinned-original-source.tex, source lines 1086–1086. Navigation pointer only: the counted text refers to the Stacks item lemma-image-irreducible-space, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-irreducible: sources/stacks-topology/pinned-original-source.tex, source lines 1103–1103. Navigation pointer only: the counted text refers to the Stacks item lemma-irreducible, which is an existing labeled result

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Source reference. Original lemma lemma-irreducible-on-top: sources/stacks-topology/pinned-original-source.tex, source lines 1332–1332. Navigation pointer only: the counted text refers to the Stacks item lemma-irreducible-on-top, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces. Assume that (a) f is open, and (b) for every $y \in Y$ the fibre $f^{-1}(y)$ is irreducible. Then f induces a bijection between irreducible components.*

Proof. We point out that assumption (b) implies that f is surjective (see Definition definition-irreducible-components). Let $T \subset Y$ be an irreducible component. Note that T is closed, see Lemma lemma-irreducible. The lemma follows if we show that $f^{-1}(T)$ is irreducible because any irreducible subset of X maps into an irreducible component of Y by Lemma lemma-image-irreducible-space. Note that $f^{-1}(T) \to T$ satisfies the assumptions of Lemma lemma-irreducible-on-top. Hence we win. □